

TOPIC 30/60

What is Supervised Fine-Tuning?

SFT: The crucial step that turns a wild text-completing calculator into a helpful, conversational AI.

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THE WILD BASE MODEL

A raw "Base Model" (fresh out of pre-training) knows the internet, but it does not know how to act like an assistant. It only completes patterns.

The Behavior:

User: "Write a poem about rain."

Base LLM: "Write a poem about snow. Write a poem about wind."

It completed your sentence template instead of answering your instruction.

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ENTER SFT



Behavior Alignment

Supervised Fine-Tuning (SFT) fixes this pattern loop. We feed the model thousands of high-quality, human-validated examples of high-tier conversations.

Instead of learning *grammar* or *everything about the world*, SFT teaches the model how to **format its outputs** and **obey user prompts**.

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THE TRAINING DATA

The Dataset

SFT requires carefully structured high-quality instruction-response datasets.

```
{  
  "prompt": "Explain gravity to a 10yo.",  
  "response": "Gravity is like an invisible glue that keeps  
our feet on the ground..."  
}
```

Thousands of clean, annotated pairs like this form the core curriculum of the SFT phase.

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UNDER THE HOOD

Masked Loss

How does the model only learn the answers without repeating the prompts?

Prompt: [Masked - No Loss calculated]

→ Response: [Calculate Loss & Updates]

The math ignores prediction errors on the user's prompt, adjusting neural weights **only** to perfect the assistant's response.

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THE LIMA SHIFT

Less is More

For a long time, companies thought SFT needed millions of conversational datasets. Then came the **LIMA hypothesis** (Meta research).

1,000 High-Quality Examples

Just 1,000 meticulously clean prompts matched with brilliant responses out-performed models aligned on 50,000 messy crowdsourced samples.

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WHAT SFT TEACHES

STRUCTURE & OUTPUT FORMAT

Reliable JSON/Markdown

SFT trains the model to structure code cleanly or output exact JSON fields required for developers.

TONE & BEHAVIOR

Helpful & Polite Persona

It stops the model from insulting the user, adopting an eager, conversational, and polite assistant tone.

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THE LIMITATIONS

Limits of SFT

While incredibly powerful, Supervised Fine-Tuning isn't a silver bullet for everything.

✗ **Sycophancy:** It learns to mimic human-validated text, often lying to agree with whatever the user says.

✗ **Hallucination:** It cannot reliably learn massive new raw facts through SFT—doing so leads to confidently wrong outputs.

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PRE-TRAINING VS SFT

Pre-training (Phase 1)

Billions of tokens. Learn grammar, code, history, and physical facts. High computational demand. Out: Raw Base Model.

SFT / Alignment (Phase 2)

Thousands of clean chat records. Learn how to listen, format outputs, and serve the user. Low compute. Out: Chat/Instruct Model.

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SFT

CONCEPT UNLOCKED



Turns text predictors into conversational agents



Emphasizes data quality over extreme volume



Restricts learning updates only to response vectors



Standardizes formatting like JSON & Markdown

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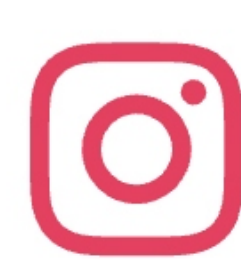
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